

 Belgium — Monthly Intelligence Report

v4.2

Edition 005 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

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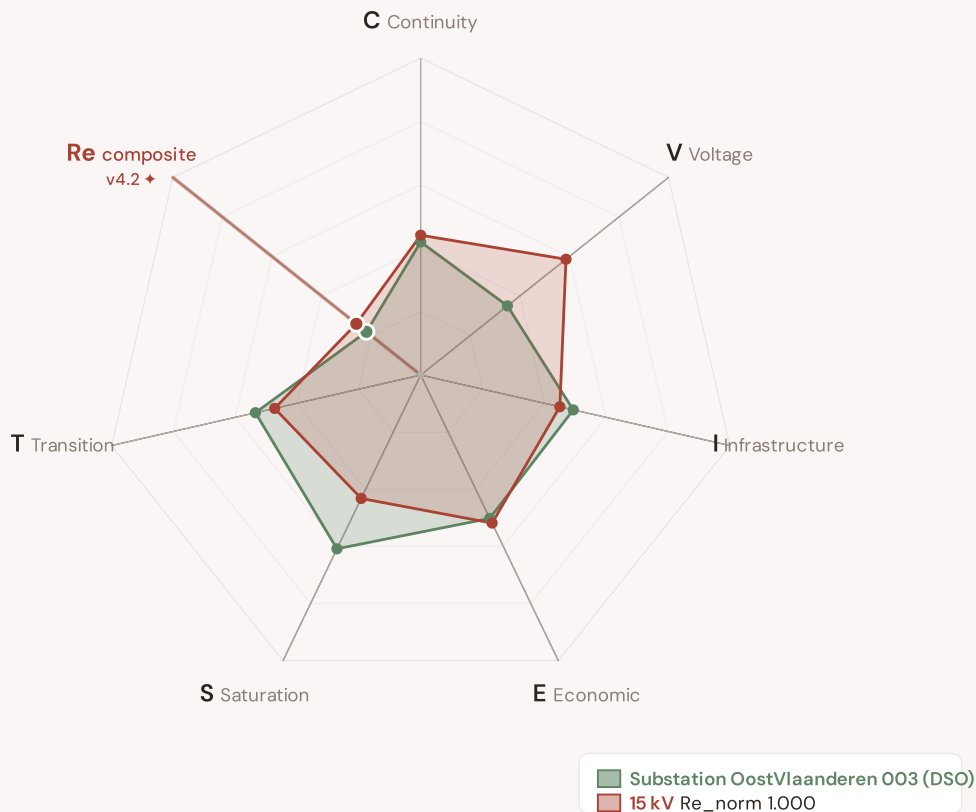
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SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). v4.2 adds **Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)). Two worked examples overlaid: **Substation OostVlaanderen 003 (DSO)** (low-exposure reference, Re_norm 0.000) · **15 kV** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

1,220

55 EHV · 199 HV · 966 distribution-tier

GRID LINES MAPPED

4,017

~5,917 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 1,220 substations and 4,017 grid lines spanning ~5,917 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Belgium's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation VlaamsBrabant 021 (MV)

Flemish Brabant, Flemish Brabant · distribution-tier

0.720

High

0.175

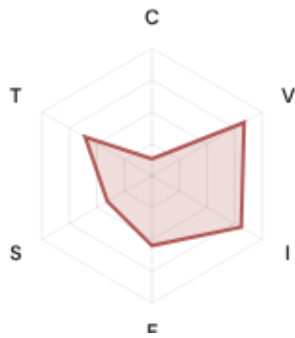
68th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



Re composite v4.2 7th axis

Re_raw 1.1398 Re_norm 0.866

ELEVATED EXPOSURE

COMPONENTS

- C** Continuity: 0.132 / 0.30 (44%)
- I** Infrastructure: 0.810 / 0.25 (324%)
- S** Saturation: 0.405 / 0.20 (202%)
- V** Voltage: 0.836 / 0.10 (836%)
- E** Economic: 0.549 / 0.10 (549%)
- T** Transition: 0.606 / 0.05 (1212%)

MODIFIERS

- R3 Consequence:** 1.077 ↑ amplifies
- R4 Graph Criticality:** 1.238 ↑ amplifies
- R6a Restoration:** 0.949 ↓ dampens
- R6b Seismic:** 0.998 → neutral
- R7 Digital Readiness:** 1.000 → neutral

This month's spotlight — automatically selected for July 2026 — is **Substation VlaamsBrabant 021 (MV)** in Flemish Brabant, Flemish Brabant. With an R_final of 0.720 (High band, 68th fleet percentile), its risk profile is dominated by the T (Transition) component at 1212% of its weight ceiling, followed by V (Voltage) at 836%. Modifiers collectively shift R_base by 46.5%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

BLIND SPOTS

0

AVG R7 MODIFIER

1.0000

HIGH-CONSEQUENCE SUBS

310

0.0% of fleet

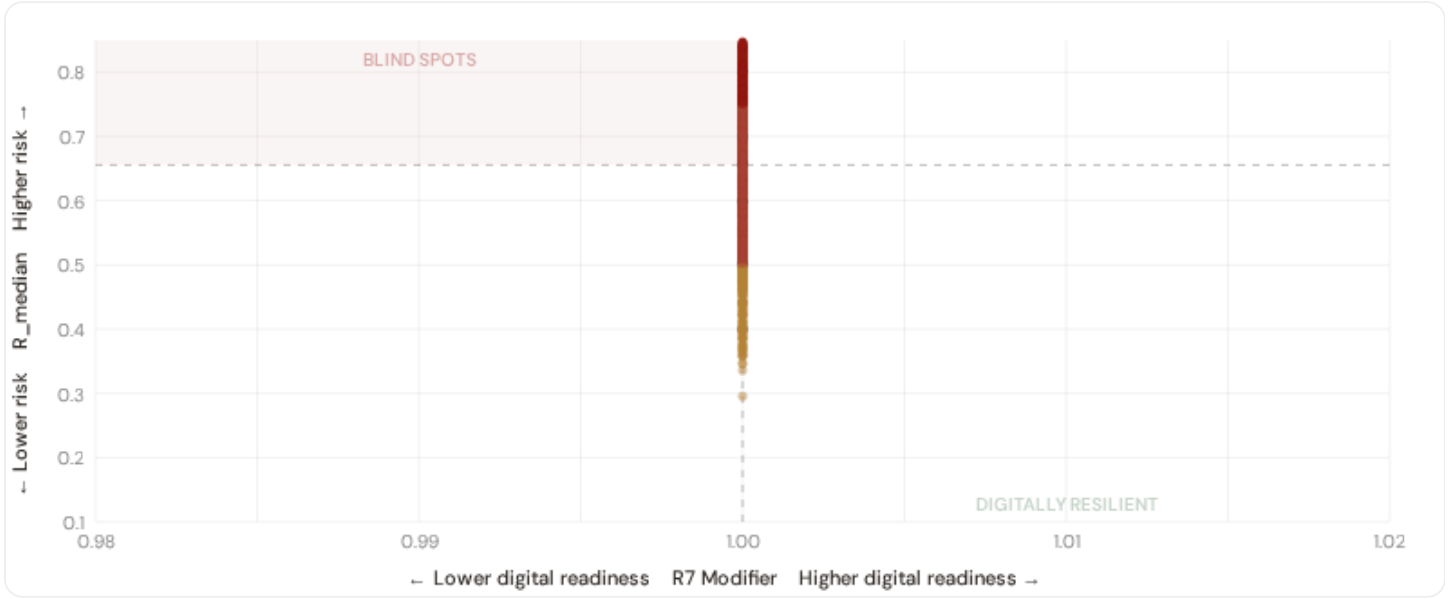
High/Critical + R7 < median

Range [0.99, 1.05]

25.4% · R3 ≥ 1.09

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in —. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh (EUR)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

310 substations (25.4%) High/Critical: **260** (83.9%) Avg R: **0.629** Avg E: **0.498** / 0.10

Low 0

Med 50

High 208

Crit 52

Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh (EUR)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

304 substations (24.9%) High/Critical: **257** (84.5%) Avg R: **0.638** Avg E: **0.494** / 0.10

Low 0

Med 47

High 199

Crit 58

Commercial / SME-Dense

Est. VoLL: €3–8/kWh (EUR)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

294 substations (24.1%) High/Critical: 247 (84.0%) Avg R: 0.655 Avg E: 0.507 / 0.10



Light / Agricultural / Rural

Est. VoLL: €1–3/kWh (EUR)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

311 substations (25.5%) High/Critical: 274 (88.1%) Avg R: 0.704 Avg E: 0.498 / 0.10

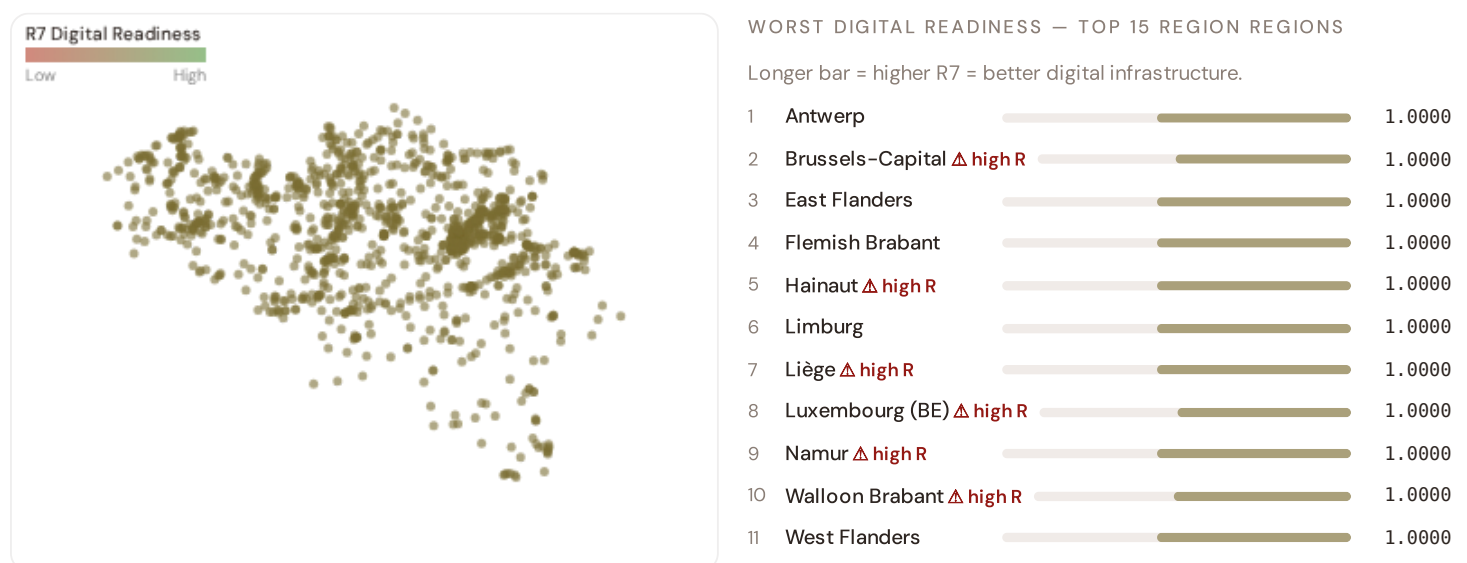


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **260 substations** combine capital-intensive economic fabric ($R3 \geq 1.09$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Antwerp (82)**, **Flemish Brabant (77)**, **East Flanders (58)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 9.8%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Antwerp** has the lowest average R7 (1.0000), while **West Flanders** leads at 1.0000 — a spread of 0.0000 across the R7 range. **0 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally

narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

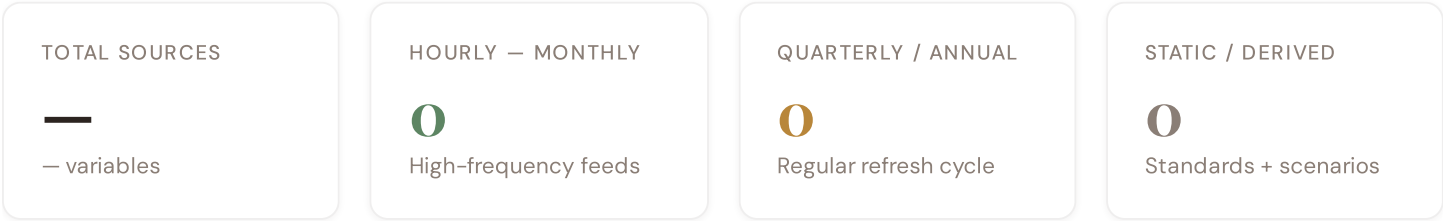
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (July 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0000, median = 1.0000; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — July 2026

OSM	OpenStreetMap Power Infrastructure	CONTINUOUS	Node / edge	4 vars
SYNERGRID	Synergrid — Federated DSO aggregate (11 DSOs)	QUARTERLY	National	3 vars
NUCLEAR	Engie + EDF Belgium — nuclear phase-out tracker	QUARTERLY	NPP	2 vars
NBB	NBB — Banque nationale de Belgique	QUARTERLY	NUTS2	2 vars
CREG	CREG — Federal Quality of Supply Reports	ANNUAL	National	5 vars
VREG	VREG — Flemish regional regulator	ANNUAL	Region	3 vars
CWAPE	CWaPE — Walloon regional regulator	ANNUAL	Region	3 vars
BRUGEL	BRUGEL — Brussels regional regulator	ANNUAL	Region	2 vars
FLUVIUS	Fluvius — Flemish single-DSO annual report	ANNUAL	DSO area	4 vars
ORES	ORES — Walloon dominant DSO annual report	ANNUAL	DSO area	4 vars
SIBELGA	Sibelga — Brussels-Capital DSO annual report	ANNUAL	DSO area	4 vars
STATBEL	STATBEL — Statistics Belgium	ANNUAL	Commune	8 vars
EUR	Eurostat — Energy Statistics (BE NUTS2 cuts)	ANNUAL	NUTS2	3 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km	2 vars
DESI	DESI Digital Economy & Society Index	ANNUAL	EU Regional	1 var
GSB	GSB — Geological Survey of Belgium + EMSC	STATIC	~5 km	2 vars
AGE-FL00D	Administration générale Eau — Flood maps	STATIC	~250 m	1 var
CATTENOM	Cattenom NPP (EDF FR) — cross-border proximity	STATIC	Province	1 var
IEEE-1	IEEE C57.91 Thermal Model	STATIC	Asset	1 var
IEC-1	IEC 60076 Power Transformers	STATIC	Asset	1 var
IEC-826	IEC 60826 Overhead Line Loading	STATIC	Zone	1 var
ISO-9223	ISO 9223 Atmospheric Corrosion	DERIVED	Province	1 var
ELIA	Elia — TSO annual + 10-min telemetry	ANNUAL + 10-MIN	National + zone	8 vars
EPEX	EPEX SPOT — BE day-ahead clearing	HOURLY	Zone	3 vars

ENTSOE	ENTSO-E Transparency — BE bidding zone	HOURLY	Bidding zone	4 vars
RMI	RMI / IRM — Royal Meteorological Institute	HOURLY + 30Y NORMS	Station	5 vars
CDS	Copernicus CDS / ERA5 + CMIP6 SSP2-4.5	HOURLY + STATIC	0.25° (~25 km)	4 vars
OPM	Open-Meteo Weather API	HOURLY	~1 km	1 var
ENISA	ENISA — EU cybersecurity index	BIENNIAL	EU country	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE

1,219

substations scored

MEDIAN R

0.655

P5=0.428 · P95=0.910

HIGH + CRITICAL

1038

85.2% of fleet

HIGH CONFIDENCE

99%

1202 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,219 substation scores remain unchanged** this edition. The fleet median R of 0.655 (mean 0.656) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 14.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per national flood-hazard tier convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	§5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	§5

V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	\$5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	\$5
V42-7	NEW	Re composite — bounded resilience scalar $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_norm = clip((Re_raw - 0.920)/(1.787 - 0.920), 0, 1)$	\$7
V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	\$14
V42-9	NEW	W1-W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 \$5.6)	\$4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 1,220 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

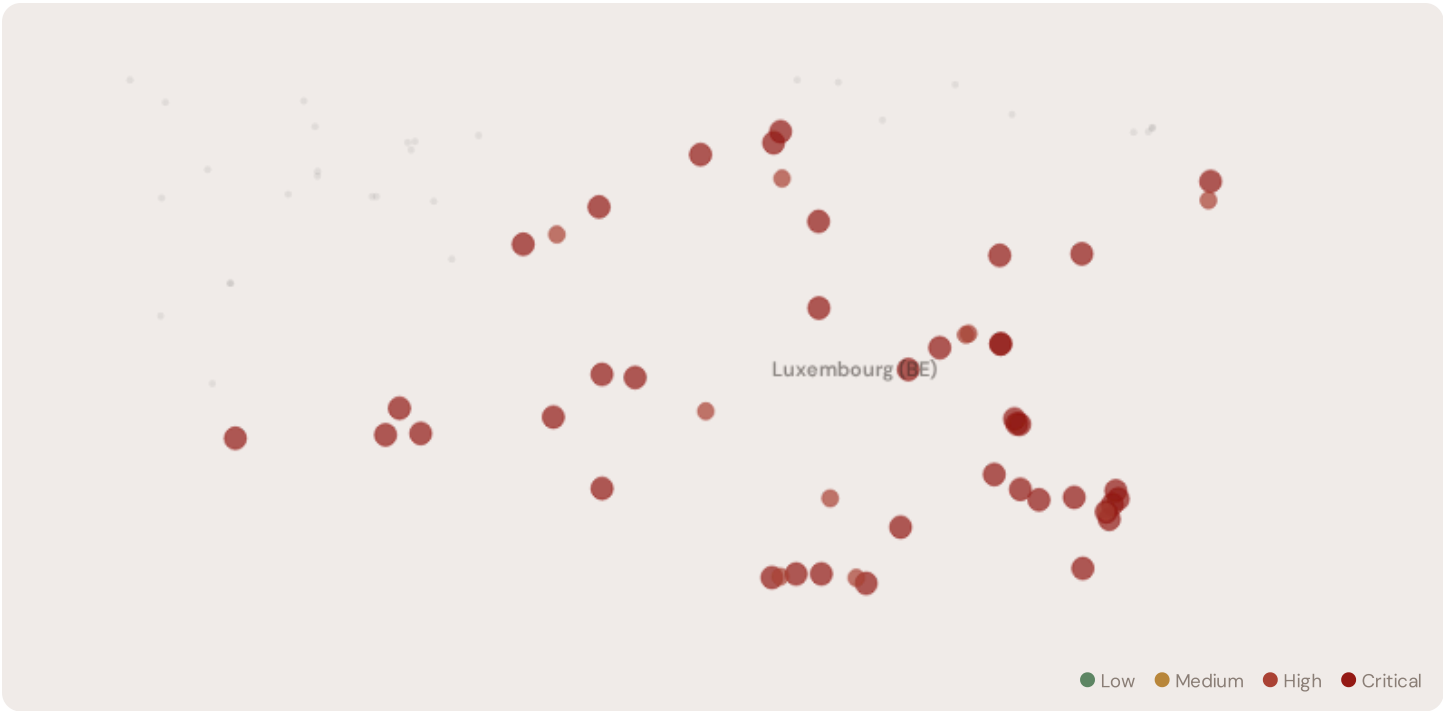
Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS 50 10 EHV · 40 HV	HIGH BAND 10 + 40 100.0% of corridor	CORRIDOR MEDIAN R 0.872 Fleet median: 0.655	WORST PROVINCE Luxembourg (BE) Avg R: 0.848 · 50 substations
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The Luxembourg (BE) hosts **50 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **10 substations (20.0%)** are classified High and **40** Critical, with only **0** in the Low band. 98% of corridor substations score above the national fleet median of 0.655. The corridor median R of 0.872 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Substation LuxembourgBE 018 (MV)	Luxembourg (BE)	0.988	Critical	1.000	0.000	0.810	0.292	0.913	0.555	C	
Poste 220/70/15 kV de Heinsch	Luxembourg (BE)	0.982	Critical	1.000	0.113	0.666	0.762	1.000	0.674	C	
Substation LuxembourgBE 016 (MV)	Luxembourg (BE)	0.975	Critical	0.963	0.138	0.670	0.483	0.988	0.423	C	
Rouvroy	Luxembourg (BE)	0.975	Critical	0.909	0.165	0.431	0.931	0.959	1.000	C	
Substation LuxembourgBE 004 (DSO)	Luxembourg (BE)	0.973	Critical	0.874	0.087	0.812	0.652	1.000	0.365	C	
Substation LuxembourgBE 008 (DSO)	Luxembourg (BE)	0.972	Critical	0.881	0.135	0.691	0.976	0.969	0.726	C	
Bonnert	Luxembourg (BE)	0.970	Critical	0.900	0.096	0.836	0.531	0.854	0.721	C	
Substation LuxembourgBE 020 (MV)	Luxembourg (BE)	0.966	Critical	0.917	0.000	0.703	0.715	1.000	0.461	C	
Substation LuxembourgBE 005 (MV)	Luxembourg (BE)	0.965	Critical	0.857	0.000	0.653	0.754	1.000	0.966	C	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Substation LuxembourgBE 009 (MV)	Luxembourg (BE)	0.950	Critical	0.964	0.134	0.617	0.683	0.903	0.458	C	

... 40 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — LUXEMBOURG (BE) VS FLEET			MODIFIER AVERAGES — LUXEMBOURG (BE) VS FLEET		
C — Continuity	0.915 vs 0.341 (+168%)		R3 Consequence	1.038	fleet 1.078 ↑
I — Infrastructure	0.349 vs 0.500 (−30%)		R4 Graph Crit.	1.238	fleet 1.238 ↑
S — Saturation	0.933 vs 0.508 (+84%)		R6a Restoration	0.949	fleet 0.949 ↓
V — Voltage	0.087 vs 0.525 (−83%)		R6b Seismic	0.996	fleet 0.997 →
E — Economic	0.571 vs 0.499 (+15%)		R7 Digital Read.	1.000	fleet 1.000 →
T — Transition	0.492 vs 0.494 (−0%)				

The dominant risk driver across the Luxembourg (BE) corridor is **T (Transition)**, averaging 0.492 against a fleet average of 0.494 — occupying 984% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.571 vs fleet 0.499 (571% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.038 (fleet: 1.078), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.996, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Luxembourg (BE)

50 subs · avg R = 0.848 · H:10 C:40 M:0

The region regions-level breakdown reveals significant intra-regional variation. **Luxembourg (BE)** leads with a mean R of 0.848 across 50 substations, while **Luxembourg (BE)** scores 0.848 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Luxembourg (BE) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

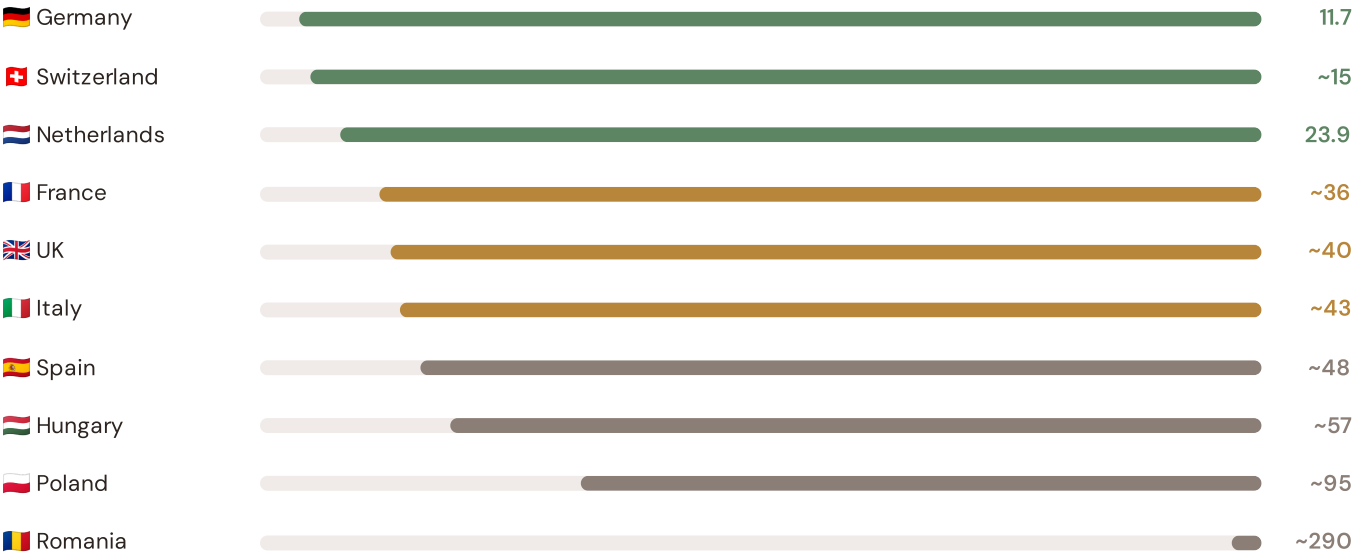
49 Luxembourg (BE) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Luxembourg (BE) sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Belgium's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



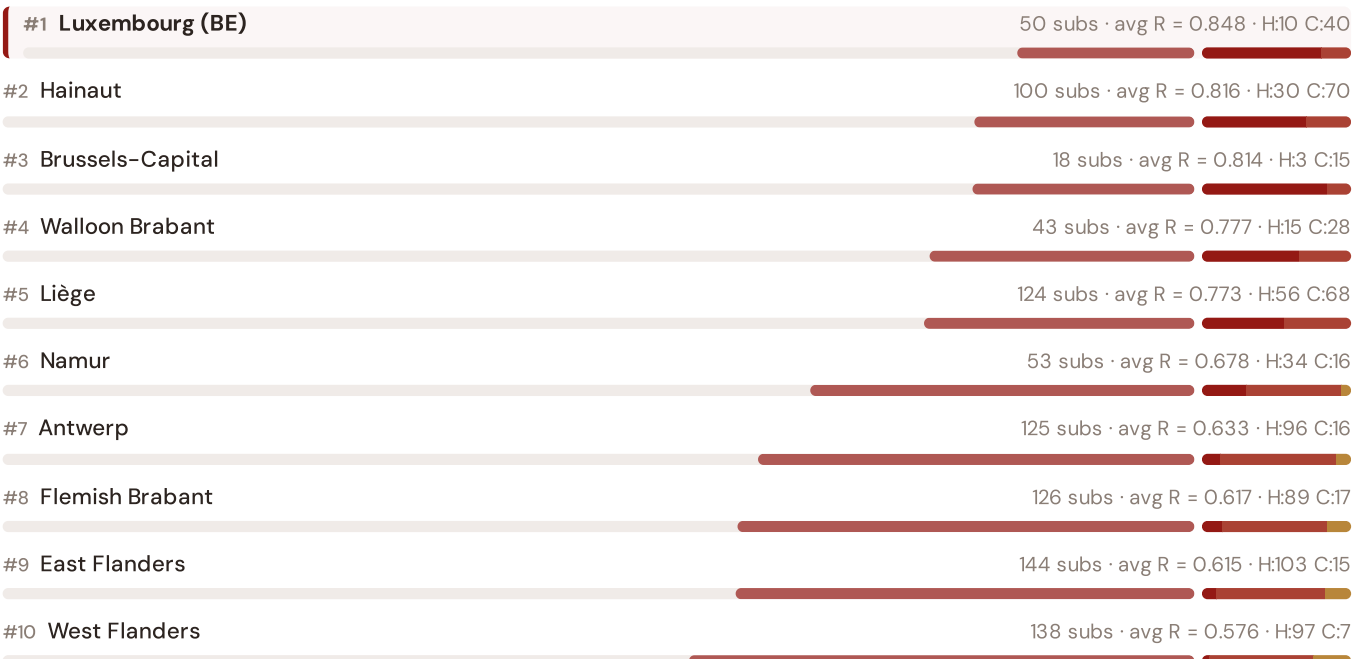
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

This month's key insight: The Luxembourg (BE) corridor deep-dive shows **50 substations** (of 50) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.915 is **2.7× the fleet average** of 0.341. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regions Risk Ranking — All 11 Belgian Regions

Where does this corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Luxembourg (BE) ranks **#1 of 11 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Luxembourg (BE), Hainaut, Brussels–Capital, while the three most resilient are Limburg, West Flanders, East Flanders. 6 of 11 score above the fleet average of 0.656. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

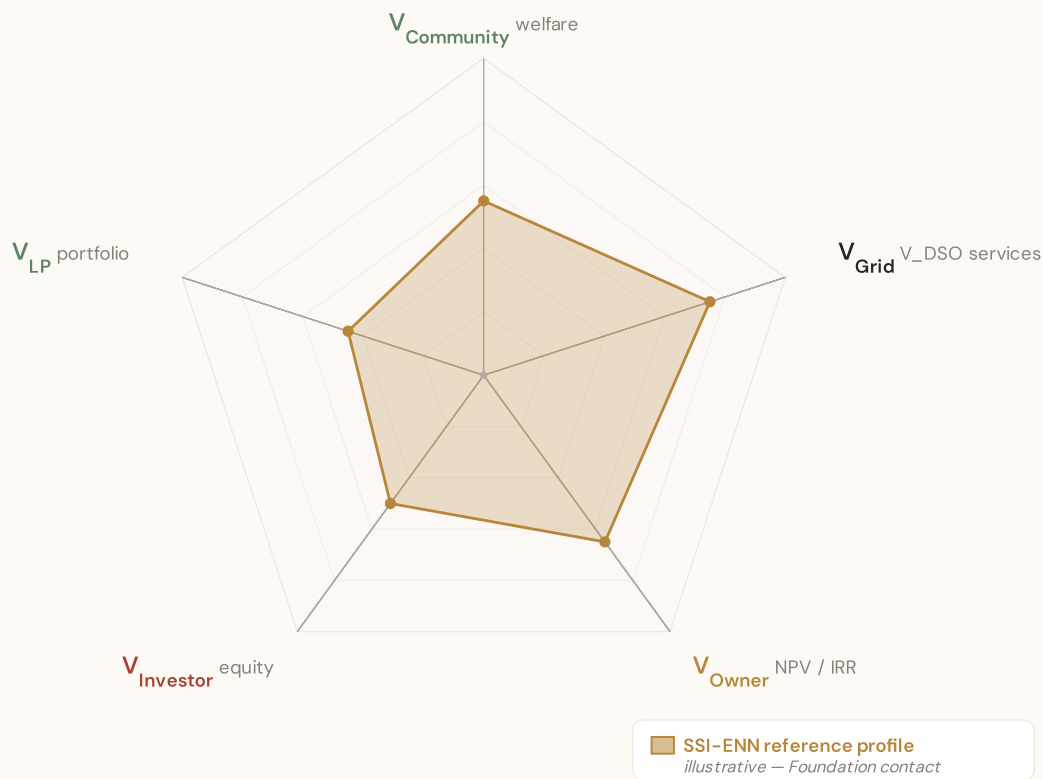
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

- Quantile regression (pinball loss) + conformal prediction
- Calibrated against SSI Monte Carlo ground truth
- Per-country, per-voltage, per-band calibration verification
- Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 1,219 substations with complete feature vectors. Average CI_width of 0.162 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 181 Medium, 731 High, 307 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 2** Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill
W1 · W2 · W3 · W4 · W8
Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker
W9 — Supply Security
Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill
W5 — Digital Inclusion
Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded
W6 · W7 · W10
Commercial deliverables — please contact the Foundation.

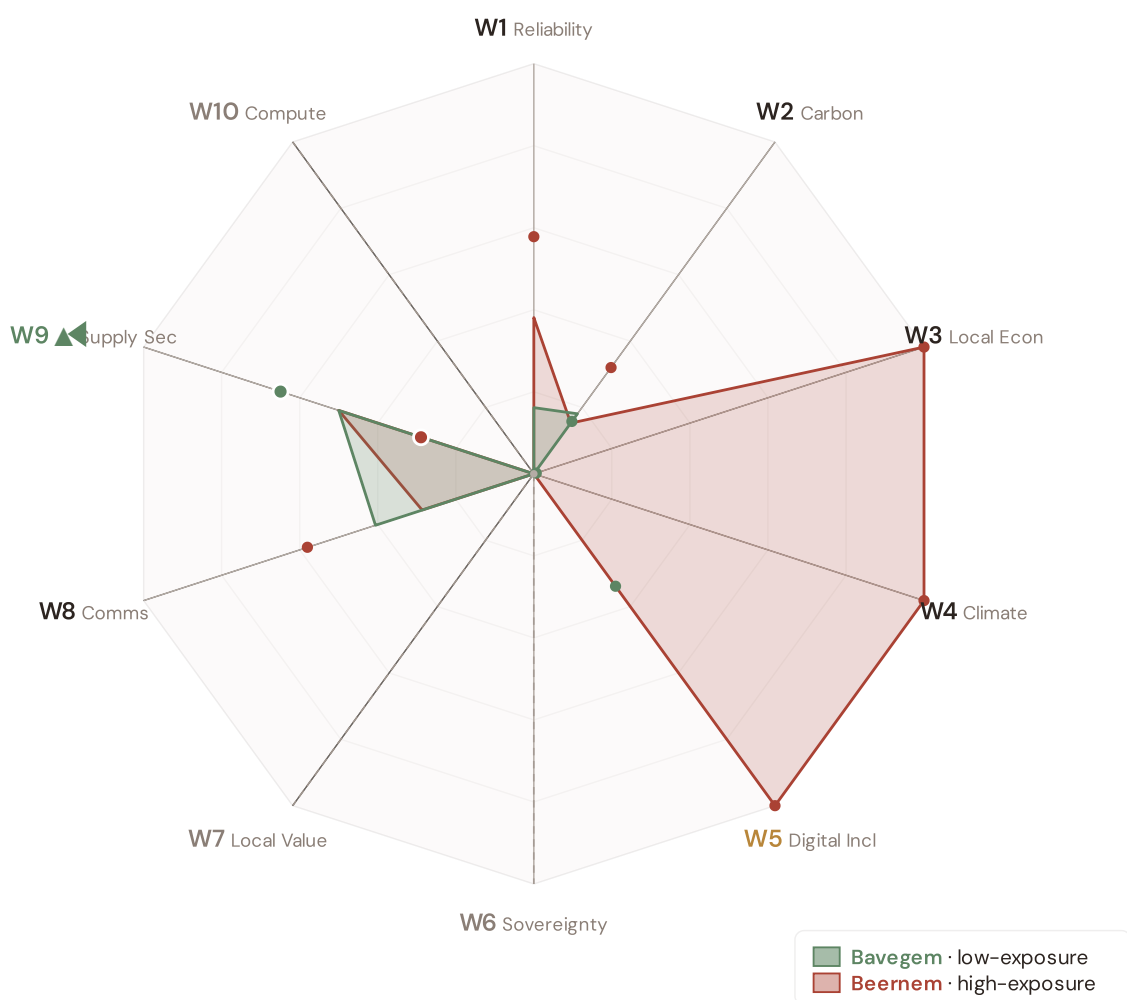
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Belgium, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to **[0, 1]** via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Substation OostVlaanderen 003 (DSO)** (low-exposure reference, Re_norm 0.000) and **15 kV** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Bavegem polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines.

Beernem polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} \text{ (SAIDI + + -POV)}$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) \text{ (+ EEA)}$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - \text{v4.2 resilience modifier family}$
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM} + + \text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population} \text{ (Markov 10-yr critical probability)}$
W9 ▲	Supply Security	Published-baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) - \text{OSM power-graph betweenness centrality} + \text{bridge identification} + \text{degree-based redundancy}$
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1–W10 per-substation baseline values for two anchor cases — Bavegem (low-exposure reference) and Beernem (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Capital Metropolitan

Deep-dive into Region 02's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Capital-intensive tier + flood overlay.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **STATBEL — Statistics Belgium** (8 variables → P1 pop density, EP_rate ~7.8% national, elderly, healthcare; per-province + per-commune).

FLEET SPOTLIGHT

o Bridge Substations — Single Points of Failure

O substations are topological bridges — their failure disconnects part of the network. Of these, **O** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Region 02**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Belgian utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 1,220 substations · 4,017 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.

